Σ^0 Production in the NuMI Beam at MicroBooNE: Analysis Note

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Abstract

Working note for the MicroBooNE Σ^0 production analysis in the NuMI beam. The selection reconstructs the $\Sigma^0 \to \Lambda\gamma$, $\Lambda \to p\pi^-$ chain in $\bar{\nu}_\mu$ charged-current interactions and is finished with a cross-validated gradient BDT. This note records the current state, the planned port into the `xsec_analyzer` framework, and the three pieces of work that follow: a Poisson-thrown fake-data sample, extension from RHC-only to the combined FHC+RHC exposure, and the selection work needed to lift a purity that currently sits near 0.4%. Revised 2026-10-15: the signal sample has been replaced by a locally produced run3b-EXT NuWro overlay, it now carries the old→current NuMI flux weight, the BDT has been retrained on 3.4× the signal statistics, and the disagreement between the available signal samples on the absolute Σ^0 rate is traced to hyperon final-state interactions (Sec. 8).

Contents

1 Status and scope **2**

2 Framework port **3**

3 Selection **3**
3.1 Signal definition . . . 3
3.2 Fiducial volume . . . 3
3.3 Candidate reconstruction . . . 3
3.4 Cut chain . . . 4
3.5 Derived observables . . . 4
3.6 BDT . . . 4

4 Why the purity is low, and what to do about it **5**
4.1 The photon candidate: what the data says . . . 5
4.2 Methods worth adopting from the single-photon analyses . . . 7
4.3 Plan: dedicated BDTs in place of one large BDT . . . 8
4.4 Proposed selection work, in order . . . 9

5 Reconstruction study on HyperonProduction ntuples **10**
5.1 Why the PeLEE ntuples cannot answer these questions . . . 10
5.2 Signal definition used in this section . . . 10
5.3 Per-particle reconstruction . . . 11
5.4 Track/shower misclassification . . . 11
5.5 The pion candidate, and why ranking cannot fix it . . . 11
5.6 The Λ displaced vertex is not usable . . . 12

6 A global hypothesis fit **12**
6.1 Motivation . . . 12
6.2 Phase 0: measured ingredients . . . 13
6.3 The mass-sculpting risk . . . 13

7 Fake data by Poisson throw **14**

8 The signal sample and its normalisation **14**
8.1 The run3b production . . . 14
8.2 The NuMI flux weight . . . 15
8.3 Why the samples disagree on the absolute Σ^0 rate . . . 15

9 Exposure: RHC now, combined later **16**

10 Work plan **17**

1 Status and scope

The analysis measures, or more realistically sets a limit on, Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions. The signature is

$$\bar{\nu}_\mu + \mathrm{Ar} \to \mu^+ + \Sigma^0 + X, \qquad \Sigma^0 \to \Lambda\gamma, \qquad \Lambda \to p\,\pi^-, \tag{1}$$

so the reconstructed final state is a muon, a proton, a pion and a photon.

The existing selection (`Sigma/pelee/SigmaSelection.C`) runs on standard PeLEE ntuples. Table 1 gives its performance scaled to the full RHC data exposure of 1.1526×10^{21} POT, before and after the 2026-10-15 change of signal sample. Efficiency is quoted against true Σ^0 in the fiducial volume.

Table 1: Selection performance, RHC only. Both rows use the same code and the same background; only the signal sample and the BDT training differ.

Stage	Raw signal	Eff. [%]	S	B	Purity [%]	$S/\sqrt{S+B}$
*Retired Run 1-EXT sample, previous BDT (*BDTG > 0.562*)*						
Preselection (NTracksCut)	3226	50.1	10.883	20618	0.053	0.0758
BDTG	1245	19.3	4.200	778	0.537	0.1502
*run3b sample, flux-weighted, retrained BDT (*BDTG > 0.616*)*						
Preselection (NTracksCut)	10931	50.8	6.370	20614	0.031	0.0444
BDTG	4904	22.8	2.850	750	0.379	0.1039

Reading the table. Two changes separate these rows from the 650/502-event numbers quoted in earlier versions of this note. The proton and pion *requirements* were made optional and are off by default – measured on this chain they remove more signal than background ($-31.6\%/-18.3\%$ and $-52.9\%/-43.5\%$) and halve the figure of merit ($0.0730 \to 0.0346$) – which roughly tripled the preselection efficiency. And the signal sample was replaced (Sec. 8).

S falls by $1.7\times$ between the two halves of the table while B is unchanged, and that is normalisation alone: $\times 0.677$ for the NuMI flux weight and $\times 0.864$ for the lower Σ^0 rate of the new production. The selection did not get worse – efficiency is unchanged at preselection and better at the operating point. Purity therefore falls for the same reason, and the honest statement is that the *previous* purity was inflated by a signal normalisation now known to be too high.
Efficiency is respectable; purity is not. At 0.379% there are $\sim$750 background events under $\sim$2.9 signal events at full RHC POT. **Improving the selection is therefore the priority**, ahead

of any normalisation or extraction work: at this purity no downstream treatment can rescue the measurement.

What this analysis produces. Not a differential cross section. At 0.1% purity and 12% efficiency the deliverable is a cross-section *upper limit*. This matters for the framework port (Sec. 2), because `xsec_analyzer` is built end-to-end for differential extraction and contains no limit machinery.

2 Framework port

The selection is being ported into `xsec_analyzer` as a `SelectionBase` subclass, alongside the CC1μ1π selections. The port buys the framework's systematic machinery – flux (PPFX multisims), cross-section (GENIE multisims and unisims), detector variations, hadron reinteraction – which is exactly the covariance a limit needs for its background prediction, and which would otherwise have to be built from scratch.

What is ported. Selection, cut flow, truth categorisation, reco and true observables, and the systematic universes at a *single* analysis bin.

What is not. Limit setting, which stays in `CrossSection.C` reading the framework covariance. An unfolded Σ^0 spectrum would be meaningless at this purity, so no multi-bin scheme is planned.

Status, 2026-10-15: not started. No `Sigma0` class exists in `src/selections`, there is no registration branch in `SelectionFactory.cxx`, and no weights have been copied into `booster_decision_tree/`. Of the nine branches below only `trk_pid_chipi_v` has since been bound, and that was for the CC1μ1π NewPID work rather than for this.

Framework gaps. Three, all small. Nine branches the selection uses are present in the ntuples but unbound in `Branches.hh` (`trk_pid_chipi_v`, `shr_energy_y_v`, `shr_p{x,y,z}_v`, `pi0_energy{1,2}_Y`, `pi0_gammadot`, `shr_moliere_avg_v`); the fiducial volume is looser than the CCπ one and must be set per selection rather than inherited; and the factory needs a registration branch. Details in `SIGMA0_FRAMEWORK_PLAN.md`.

3 Selection

3.1 Signal definition

`nu_pdg` $= -14$, `ccnc` $= 0$, a Σ^0 (PDG 3212) in the final-state list `mc_pdg`, and the true vertex inside the fiducial volume. Photons are not stored in `mc_pdg`, so the decay is tagged by the Σ^0 itself rather than by its daughters.

3.2 Fiducial volume

$3.00 < x < 253.35$, $-112.53 < y < 114.47$, $3.1 < z < 1033.0$ cm, excluding the dead region $675.1 < z < 775.1$ cm. This is deliberately looser than the CC1μ1π box (10–246, $\pm$101, 10–986): the signal is rare enough that acceptance is worth more than vertex quality.

3.3 Candidate reconstruction

- **Muon**: longest track with `trk_score` > 0.8, start within 20 cm of the vertex, length > 10 cm, and LLR PID > 0.7.

- **Proton and pion**: assigned jointly by the combined discriminant

$$D = \frac{\mathrm{LLR}}{0.591} + \frac{\chi^2_p - \chi^2_\pi}{124.5}, \tag{2}$$

with proton $= \arg\min D$ and pion $= \arg\max D$. This separates true protons from true pions with AUC 0.873, against 0.838 for χ^2_p alone and 0.852 for LLR alone. The χ^2 term is dropped per track when undefined ($\sim$30% of tracks).

- **Photon**: the two non-track PFPs closest to the vertex; the closest is taken as the Σ^0 decay photon.

3.4 Cut chain

Nine cumulative stages: `AllEvents`, `FiducialV`, `CosmicVeto` (`topological_score` > 0.20), `NShowers` ($0 < n_{\mathrm{shr}} < 4$), `MuonCandidate`, `ProtonCandidate`, `PionCandidate`, `NTracksCut` ($n_{\mathrm{trk}} < 5$), `BDTG`.

3.5 Derived observables

Λ mass from (p, π); Σ^0 mass from (Λ, γ) using the per-PFP shower energy `shr_energy_y_v`; Λ and photon vertex-pointing angles and impact parameters; Λ decay distance; and a π^0 di-photon mass built from the framework π^0 reconstruction for background rejection.

3.6 BDT

Gradient BDT over 57 inputs, evaluated with two-fold cross-validation: folds are trained on even and odd event numbers and each is applied only to the events it did not train on. The cross-validation is not optional – applying a fold to its own training events would inflate the ROC and the apparent significance.

The input list grew from 49 with the Second-Shower-Veto block of Sec. 4.2 (unassociated-hit count and fraction, recomputed per event rather than taken from `slnunhits`, plus `slclustfrac`, `secondshower_Y_nhit`, `secondshower_Y_dot` and the sub-cluster counts) and the shower Molière angle.

Retrained 2026-10-15. On the run3b sample the training tree grows from 3226 to 10931 signal events and the discriminant improves measurably. Scores are cross-validated throughout, so these are out-of-sample:

	AUC	B/B_0 at fixed signal efficiency		
		50%	40%	20%
retired sample, previous weights	0.8429	0.0681	0.0414	0.0125
run3b sample, previous weights	0.8333	0.0688	0.0432	0.0132
run3b sample, retrained	**0.8656**	**0.0467**	**0.0293**	**0.0084**

At matched signal efficiency the retrained BDT rejects $\sim$32% more background. The operating point moves from BDTG > 0.562 to > 0.616, where it keeps *more* signal (44.7% of preselection against 41.4%) with *less* background (750 against 967). The gain folds together 3.4$\times$ the training signal and training on cosmics from the matching run period; separating the two would need a third training on a 3226-event subsample, which has not been run.

This lever is now close to exhausted. A learning curve on fold A, with the test sample held fixed, gives ROC 0.857, 0.872, 0.881 for 676, 1352 and 2704 training signal events. Returns halve per doubling and the curve heads for $\simeq 0.89$, so doubling the production again would buy ~ 0.005 in AUC, which by the scaling above is a few percent of background. Against MC statistical errors of 1.5% on S and 1.6% on B at the operating point, and a 2% background systematic worth ± 15 events on 2.9 signal, **the analysis is now systematics-limited rather than signal-statistics-limited**. That is a change from every earlier version of this note.

4 Why the purity is low, and what to do about it

This is the priority. Two diagnostics point at the same place.

Table 2: Surviving background by interaction mode. Measured at the looser operating point in use when this was written, so the absolute yields are not those of Table 1; the fractions are what carry over, and the background itself is unchanged by the 2026-10-15 signal replacement.

Mode	Scaled yield	% of background
Resonant (RES)	1381.8	52.3
Deep inelastic (DIS)	635.3	24.0
Quasi-elastic (QE)	363.8	13.8
Meson exchange (MEC)	177.1	6.7
Cosmic	74.6	2.8
Coherent (COH)	8.9	0.3
Total	2643.1	100

The surviving background is π^0 events. RES and DIS together are 76%. Both produce $\pi^0 \to \gamma\gamma$, which supplies a fake Σ^0 photon, plus charged hadrons that fake the proton and pion tracks. The background is not diffuse: it is one physics process.

The candidate assignment is the weak link. Truth-matched purities of the individual candidates in background events:

Candidate	Correct [%]	Dominant impostor
Muon	81	—
Proton	58	—
Pion	31	another proton (28%)
Photon	29	a proton (18%)

The muon is reliable and the proton is acceptable, but **the pion and photon candidates are wrong roughly 70% of the time**. Since both the Λ and Σ^0 masses are built from them, the two most powerful BDT inputs are being computed from misidentified objects in most background events – and, importantly, we did not know how often in signal events, because the PeLEE ntuples carry no truth for the decay products at all. That gap is what Sec. 5 closes on the hypana ntuples; the signal sample is in any case now overlaid on real EXT data rather than cosmic-free.

4.1 The photon candidate: what the data says

This was investigated first, and the result contradicts the obvious expectation, so it is recorded in full.

Composition. Truth-matching the photon candidate in background events (69557 preselected events) gives:

Truth match	Fraction [%]
photon	30.0
unmatched (PDG 0)	26.7
proton	15.2
charged pion	16.3
muon	9.8
electron	1.4

Charged tracks reconstructed as showers – proton, pion, muon together – account for 41%, and a further 27% has no truth match at all.

The signal photon is soft, which inverts the strategy. The Σ^0–Λ mass splitting is only 77 MeV, so

$$E_\gamma = \frac{m_{\Sigma^0}^2 - m_\Lambda^2}{2m_{\Sigma^0}} = 74.5 \text{ MeV} \tag{3}$$

in the Σ^0 rest frame. The measured mean energy of the photon candidate in signal events is 70 MeV, which matches that prediction and is good evidence that the selection is finding the right object. The background photon candidate averages 140 MeV, because the π^0 photons that dominate it are far harder.

Requiring shower-like quality would therefore remove signal preferentially, which is the opposite of what an earlier draft of this note proposed. A 74 MeV photon is near the reconstruction threshold: it makes few hits and deposits little energy. Measured against true photons in background events – which are π^0 photons and much harder – the signal photon looks *less* shower-like, not more.

Cutting in the correct direction helps, but only before the BDT. Scanning cuts in the soft direction at preselection:

Cut	ϵ_S [%]	Background rejected [%]	FoM gain
$E_\gamma < 120$ MeV	94.5	25.4	$\times$1.092
$N_{\text{hits}}^Y < 102$	95.7	21.5	$\times$1.079
$E_\gamma \in [5, 100]$ MeV	70.6	55.2	$\times$1.052

But `Shower1Energy` is already one of the 49 BDT inputs, so the question is whether the cut adds anything the BDT has not already taken. Applied on top of the BDT operating point it does not:

Selection	S	B		FoM
BDT only	502	20762		3.443
BDT + $E_\gamma < 120$ MeV	482	18663	3.484	($\times$1.012)
BDT + $E_\gamma < 100$ MeV	456	17706	3.384	($\times$0.983)

Conclusion. The photon candidate cannot be improved by cutting on variables the BDT already has – it is using them close to optimally. An explicit π^0 mass veto is likewise ineffective, rejecting only 9% of background, because it requires *two* reconstructed showers and most background events supply only one. Improvement must come from information the BDT does not yet have.

The one genuinely unused handle. `shr_moliere_avg_v`, the shower Molière angle, is in the ntuples and in the plotting variable list but is *not* among the 49 BDT inputs. It measures lateral spread, which is orthogonal to energy and hit count and is the classic discriminant between an electromagnetic shower and a track stub. Adding it requires re-running the selection to dump it into the training tree, then retraining both folds. That is the concrete next step for the photon candidate, and it should be judged by the FoM gain on top of the BDT, not at preselection.

4.2 Methods worth adopting from the single-photon analyses

The gLEE single-photon search and the NCπ^0 technote solve a problem very close to ours: isolate a single photon against a background of π^0 events in which one decay photon was missed. Four of their methods transfer directly.

1. Second-Shower Veto – look for the photon that was *not* reconstructed. This is the most important one. Our π^0 veto requires *both* decay photons to be reconstructed, which is why it fires so rarely and rejects only 9% of background (Sec. 4.1). gLEE instead trains a dedicated Second-Shower Veto (SSV) BDT on evidence that a second photon exists even when Pandora never built a shower from it. Their SSV inputs include:

- the **fraction of unassociated hits in the slice** – hits in the neutrino slice belonging to no reconstructed object, which is exactly what a missed low-energy photon leaves behind;
- the **number of unassociated hits** (used logarithmically);
- a **minimum conversion parameter**, and the implied invariant mass of the candidate photon paired with any second-shower candidate.

Applied to Σ^0, this attacks the dominant background where it actually lives: our RES and DIS background is 76% of the total and is π^0-driven, and the failure mode is precisely a missed second photon. This should be tried before any further work on the existing π^0 mass variable.

The ingredients are already in our ntuples, checked directly on the signal file: `slnhits` and `slnunhits` (total and unassociated hits in the slice, whose ratio is gLEE's fraction variable), `slclustfrac`, `secondshower_Y_nhit` and `secondshower_Y_dot` from the PeLEE second-shower tagger, and `shrsubclusters0/1/2` (sub-cluster counts, which the CC1μ1π BDT already uses). None of these is currently read by `SigmaSelection.C`. Adding them is the same mechanical exercise as the Molière angle – bind the branch, add to both variable lists, retrain – but on variables with a far better physics motivation.

2. One BDT per background class, not one BDT for everything. gLEE runs *five* BDTs for its 1γ1p channel – Cosmic, ν_e, BNB-Other, NCπ^0 and SSV – each trained against one background, with a separate cut on each score. We use a single BDT over 57 inputs trained against all backgrounds at once. Given that our background decomposes cleanly (RES 52%, DIS 24%, QE 14%, MEC 7%, cosmic 3%), a dedicated π^0-rejection BDT and a separate cosmic BDT would let each learn its own discriminating structure instead of averaging over processes that look nothing alike. This is an architectural change and the single largest departure from the current design.

3. Conversion distance as a first-class variable. gLEE uses the shower conversion distance throughout, in both linear and logarithmic form, as one of its most-shown discriminants. We have `Shower1VtxDist`, which is the distance from the vertex to the shower start and is closely related, but it is one input among 49 rather than a variable the selection is built around. A real photon converts a few centimetres from the vertex; a proton track begins at it. Worth examining separately and in log form, since gLEE's plots show the discrimination lives at small distances where a linear axis compresses it.

4. Constrain the π^0 rate from data. The NCπ^0 technote fits the π^0 normalisation to data and finds GENIE over-predicts non-coherent NCπ^0 by 25%: the flat fit gives $N_{\mathrm{non-coh}} = 0.75$ ($\chi^2/\mathrm{ndf} = 1.96/4$), and gLEE separately uses a flat 0.69 or a momentum-dependent $0.92 - 0.80\,|\vec{p}_{\mathrm{true}}|$ scaling. Since our background prediction is dominated by the same physics, an unconstrained π^0 normalisation is both a bias and a large systematic. A sideband constraint would reduce both. Note that gLEE deliberately does *not* apply the scaling in its final fit, relying on the covariance matrix instead – that is the more defensible route for us too, and it argues for making sure the π^0 normalisation uncertainty is properly represented in the covariance rather than for rescaling the central value.

What does not transfer. gLEE's signal is a single photon with no muon, so its ν_e and BNB-Other BDTs target backgrounds we do not have; our muon candidate is already 81% pure and is not where we are losing. Their analysis is also BNB, so the flux and interaction mix differ, and any borrowed *numerical* threshold would have to be re-derived.

4.3 Plan: dedicated BDTs in place of one large BDT

Why the single BDT is the wrong shape. One BDT over 57 inputs is trained against a background that is really four unrelated physics processes – π^0-producing RES (52%) and DIS (24%), quasi-elastic (14%) and MEC (7%) events with no π^0 at all, and cosmics (3%). A single discriminant must find one surface separating signal from *all* of them at once, so it learns whatever those backgrounds have in common and averages away the structure specific to each. gLEE runs five, and their signal is rarer than ours.

The evidence that this is the binding constraint rather than variable quality is that adding variables barely moves the answer: the Molière angle changed the figure of merit by $\times 1.013$, and the ROC plateaus regardless of background size or added inputs. A better-shaped discriminant is a different lever from a better variable.

Proposed structure. Three BDTs, cut independently, replacing the single BDTG:

BDT	Trained against	Fraction of bkg	Discriminating on
π^0-rejection	RES + DIS with a true π^0	$\sim$76%	SSV block, π^0 mass, shower shape
Non-π^0 ν	QE + MEC + COH, no π^0	$\sim$21%	track multiplicity, PID, Λ mass
Cosmic	cosmic-tagged events	$\sim$3%	topological score, containment, Λ pointing

Each is trained signal-versus-one-class, keeping the same two-fold cross-validation by event-number parity. The truth labels needed to define the classes already exist in the training dump: `Interaction`, `HasPi0`, `CosmicMu` and `Ccnc` are written to `BDTInput.root` as diagnostic branches, so the split costs no new reconstruction, only a change to how `TrainBDT.C` builds its background tree.

How to combine them. Two options, and the choice should be made on measurement rather than taste:

- **Independent cuts**, as gLEE does – one threshold per score, tuned jointly. Transparent, and each cut can be validated in its own sideband.
- **A second-stage combination** – feed the three scores into a small final discriminant. Usually stronger, but harder to validate and easy to overtrain on 650 signal events.

Given our signal statistics, independent cuts are the safer starting point.

The statistics problem this creates, and the answer to it. Splitting the background does not reduce the signal available for training: every signal event is used in every BDT. It does reduce the *background* per training, but from ∼70,000 preselected events, so even the cosmic class retains thousands. When this was written the real constraint was the 650 signal events, which is why it ended by pointing at the incoming cosmic-overlay signal samples. Those have since arrived: the training tree now holds 10931 signal events (Sec. 8), so the statistics objection to splitting has gone.

Order of work. Build the π^0-rejection BDT first and evaluate it against the current single BDT at matched signal efficiency. If it alone does not improve the figure of merit, the split is not the answer.

Status. The split was implemented — `TrainBDTSplit.C` trains the three class BDTs by fold, `apply_split.C` scores the tree into `SplitScores.root` and `eval_split.C` compares — and run once on an earlier and much smaller training tree (1004 signal events). No evaluation against the single BDT survives in a retained log, and the single BDT has since been retrained and improved substantially (AUC 0.8656), so the comparison has to be remade on the current sample before the split can be called a gain or a dead end. Redirecting the effort "to signal statistics instead" is no longer the fallback it was: that work is done.

Interaction with the SSV variables. The SSV block belongs naturally in the π^0-rejection BDT, where its targets are. Adding it to the single BDT first – as has now been done – is worth doing anyway, because it measures how much of the gain comes from the variables and how much from the restructuring: if the SSV variables give little in the single BDT but a lot in a dedicated π^0 BDT, that is direct evidence for the split.

4.4 Proposed selection work, in order

1. **Second-Shower Veto variables** (Sec. 4.2): unassociated-hit fraction and count in the slice. This targets 76% of the background at its actual failure mode and is expected to dominate everything else on this list.

2. **Split the BDT by background class** (Sec. 4.2), starting with a dedicated π^0-rejection BDT.

3. **Add `shr_moliere_avg_v` to the BDT.** *Done, and it does almost nothing*, as its separation predicted: 503 signal against 502, purity 0.111% against 0.108%, figure of merit 0.0544 against 0.0537, a gain of ×1.013. Recorded because it closes off the last unused photon-shape variable and supports the argument of Sec. 4.3 that variable quality is no longer the binding constraint.

4. **Improve the pion candidate (31% correct).** The joint discriminant of Eq. 2 is already a real improvement over χ^2_p alone, but the failure mode is specific: a second proton is taken as the pion 28% of the time. Adding the sign-sensitive information – track length against the proton range curve, or the Λ-mass constraint itself as part of the assignment rather than as a consequence of it – would target that directly.

5. **Reconsider the Λ decay topology.** Λ has $c\tau = 7.89$ cm, so a genuine $\Lambda \to p\pi^-$ has a displaced vertex. The selection computes `LambdaDecayDist` and the pointing angle but does not cut on them. A displaced-vertex requirement is close to background-free in principle and is the most distinctive feature of the signal.

Earlier versions of this note closed here by calling the analysis **signal-statistics limited** – 650 raw signal events at preselection, with a BDT ROC that plateaued regardless of background size or added variables – and naming more signal MC as the largest single lever. **That lever has now**

been pulled. The run3b production (Sec. 8) gives 10931 raw signal events at preselection, and the retrained BDT rejects 32% more background at matched signal efficiency (AUC 0.8333 → 0.8656). The plateau was therefore a property of the training sample, not of the variables.

It is also close to exhausted: the learning curve of Sec. 3.6 is flattening, doubling the sample again is worth a few percent of background, and the MC statistical errors at the operating point (1.5% on S, 1.6% on B) sit far below a 2% background systematic. The binding constraint has moved to systematics and to the candidate assignment, which is where the remaining items on this list point.

5 Reconstruction study on HyperonProduction ntuples

5.1 Why the PeLEE ntuples cannot answer these questions

Every question about *which particle a candidate actually is* was unanswerable on the PeLEE ntuples. Their truth record stops at the primary vertex: `mc_pdg` lists the Σ^0 and the muon, but never the $\Sigma^0 \to \Lambda\gamma \to p\pi^-\gamma$ chain. Measured on the signal sample, `nproton` averages 0.06 and is zero in 94.4% of events, `npion` is zero in 100%, PDG 22 never appears, and *zero of 17721 events* contain a backtracked π^- PFP. The per-candidate truth columns are correspondingly meaningless: the pion candidate matches a true π^- in 0.0% of events, 60% carry no truth match at all and 36.5% leak the muon's truth. This is a property of the ntuple, not a bug in the selection.

The HyperonProduction ("hypana") analyser writes the full decay chain (`mc_sigmazero_lambda_end_*` is the true Λ decay vertex, `mc_decay_*` the daughters) together with per-PFP truth matching (`pfp_true_pdg`, `pfp_purity`, `pfp_completeness`). The studies below are all performed there. `HypInput.h` adapts the hypana branches to the PeLEE names the selection already uses; the two formats index every `trk_*`/`shr_*`/`pfp_*` vector per PFP with equal lengths, so the port is a branch remap with a double→float copy rather than a structural rewrite.

Two caveats applied when this study was made, and one of them has since gone. The hyperon samples are generated on a single channel with a dedicated normalisation, so they are valid for efficiencies and ratios; their absolute yields are the subject of Sec. 8. The second caveat — that hypana output existed only for the hyperon samples, with none for the generic RHC overlay, and that the overlay's parent reco2 files were not on local disk — no longer holds. The RHC overlay has since been processed through reco2 and HyperonProduction: `lists/bkg_overlay.list` now holds 14,024 hypana files at 6.9701×10^{21} POT, 70% of the full RHC exposure, so a hypana-only analysis with realistic background is possible. Dirt in hypana format is still missing. The numbers below were measured on the retired NuWro sample and have not been remade on the run3b one; the measurements are efficiencies and ratios, which the swap does not move, but the statistics behind them are now 3.4× larger if any of them is worth sharpening.

5.2 Signal definition used in this section

Σ^0, $\bar{\nu}_\mu$ charged current, $\Lambda \to p\pi^-$ in truth, muon momentum above 0.15 GeV/c, and **both** the reconstructed and the true neutrino vertex inside the fiducial volume. The branching ratio matters: only $\sim$ 60% of Σ^0 events have $\Lambda \to p\pi^-$ at all (PDG 63.9%), so the remaining 40% contain no proton or pion to reconstruct and must not be counted as reconstruction failures.

A fiducial-volume trap. Requiring only the *reconstructed* vertex in the FV gives badly wrong answers, because events whose reco vertex is in the FV but whose true vertex is not are mis-reconstructed — the reco vertex belongs to a different interaction — and they dominate the sample. With reco FV only the muon appears to be reconstructed 53.7% of the time and all four particles 10.2%; requiring both fiducial cuts gives 91.5% and 18.1%. The internal check that catches this: a ceiling *below* the selection's own preselection efficiency (15.6%) is impossible.

5.3 Per-particle reconstruction

Table 3 gives, for each of the four final-state particles, how often a truth-matched PFP exists and where Pandora places it in its hierarchy. Statistics: 3704 signal events, full NuWro sample.

particle	primary (gen 2)	secondary (gen ≥ 3)	not reconstructed	total
μ^+	91.5%	0.3%	8.2%	91.8%
p	58.1%	2.8%	39.1%	60.9%
π^-	45.7%	1.6%	52.8%	**47.3**%
γ	59.8%	2.6%	37.6%	62.4%

Table 3: Reconstruction of the four Σ^0 final-state particles. The π^- is the weak link.

Found together: 0 of 4 in 4.5% of events, 1 in 7.5%, 2 in 27.2%, 3 in 42.6%, and **all four in** 18.1%. Since the selection requires a muon, proton, pion and shower candidate, 18.1% is its hard ceiling; it reports 15.6% at the `NTracksCut` preselection only because it fills those slots with whatever PFP is available.

Secondaries do not explain the loss. The analyser originally discarded every PFP with generation $\neq 2$. The natural hypothesis was that the Λ daughters, produced at a displaced vertex, were being reconstructed as generation-3 secondaries and thrown away. The module was changed to keep them (labelled by a new `pfp_generation` branch, so `pfp_generation == 2` recovers the old behaviour exactly) and the measurement repeated: secondaries are 6.6% of PFPs but recover only 2.8% for the proton and 1.6% for the pion, and nothing at all for the muon and photon. The losses are genuine reconstruction failures.

5.4 Track/shower misclassification

In events with at least four PFPs (622 events), each truth-matched particle was compared against the track/shower split the selection uses (`pfp_trk_shr_score` > 0.5):

particle	expected	as track	as shower	not found
μ^+	track	95.7%	1.6%	2.7%
p	track	70.3%	8.2%	21.5%
π^-	track	54.2%	**11.6**%	34.2%
γ	shower	**15.0**%	69.5%	15.6%

Table 4: Classification of truth-matched particles. Bold entries are misclassifications that directly corrupt candidate assignment.

All four particles are *found* in 41.2% of these events but all four are *correctly classified* in only 26.7%: the hard split at 0.5 costs 14.5 percentage points, a third of otherwise-perfect events. Two consequences matter. Photons pass as tracks 15% of the time and then compete for the pion slot — which is the origin of the 25% photon contamination measured independently among pion candidates. And π^- are called showers 11.6% of the time, which removes them from the track pool entirely so they can never be selected.

5.5 The pion candidate, and why ranking cannot fix it

On signal events reaching the pion stage the candidate is a true π^- only 34.2% of the time (muon 89.5%, photon 77.6%, proton 64.5%), with contaminants γ 25.0%, p 16.2%, μ^+ 10.5%. Two hard constraints make this not a ranking problem:

- A true π^- PFP exists in only 56.6% of these events (track-like in 52.6%). That is the ceiling: 43% of the time there is no pion to find and the selection assigns one regardless.
- 76.3% of events have exactly *one* track-like PFP left after the muon and proton are taken, so there is no choice to make. Adding wiggliness and dE/dx terms to the ranking changed the outcome by exactly zero events across seven weightings.

Raising the pion track-score threshold lifts purity from 34.2% to 41.6% but the figure of merit falls monotonically (ratio $1.000 \to 0.442$ over $0.50 \to 0.95$), so it is not worth doing.

5.6 The Λ displaced vertex is not usable

The Λ displaced-vertex handle fails on physics grounds, not implementation. Measured with truth-matched p and π^- tracks ($n = 266$):

- True Λ flight distance: median 1.85 cm, mean 3.23 cm. Not 7.89 cm — $c\tau$ is the rest-frame value and the Λ from a Σ^0 decay is slow ($\beta\gamma \approx 0.4$), so $\beta\gamma c\tau \approx 3$ cm.
- Reconstructed primary-vertex resolution: median 5.54 cm.
- Correlation between reconstructed and true flight distance: $r = -0.014$.

The ruler is three times longer than the quantity being measured. Three estimators were compared against the true decay vertex — the midpoint of the two track starts (what the code used), the distance of closest approach of the two track lines, and the best of the four endpoint pairings — and all three gave a residual of ~ 5.5 cm, because all are limited by the same vertex resolution. The tracks themselves converge well (DCA median 0.33 cm); it is the baseline that is poor. This closes the earlier null result (`LambdaDecayDist` signal 10.19 cm against background 11.02 cm) and the requirement is withdrawn from the work plan. It would still work for directly produced, faster Λ, so the negative result should not be generalised.

6 A global hypothesis fit

6.1 Motivation

The selection currently identifies the muon, then the proton, then the pion, each conditioned only on what has already been taken, and never uses the fact that $p\pi^-$ must form a Λ and $\Lambda\gamma$ a Σ^0. The particles are not independent and are being treated as if they were. The per-role PID templates in Sec. 6.2 show why this fails specifically on the pion: the muon, proton and photon each have a strong, distinctive signature, while **the π^- has none** — it lies between the other three on every variable measured. It can only be identified by exclusion once the others are pinned, together with the Λ mass constraint, which is precisely what a sequential cut chain cannot do.

The proposed replacement is:

1. **Relaxed candidate pool.** Drop `trk_score` > 0.5 as an eligibility gate; every PFP is eligible for every role and the score becomes a likelihood term. This recovers the 11.6% of π^- called showers and stops photons monopolising the pion slot.
2. **Enumerate assignments** of four roles to N PFPs, $P(N, 4)$. Computationally free: $N = 4 \to 24$, $N = 6 \to 360$, $N = 8 \to 1680$.
3. **Score each hypothesis** with a combined $-2 \ln L$ built from the two mass constraints, per-role PID templates, muon-at-vertex, the p–π DCA, a length-ordering prior and a soft-photon prior.

4. **Partial hypotheses** for the 29.1% of signal with three PFPs, where the dominant topology is 2 tracks + 1 shower (51.9%), i.e. one hadron missing with the photon still present.

The headroom is specific: all four particles are found in 41.2% of $N \geq 4$ events but correctly classified in 26.7%, so perfect assignment given what is found is worth a factor 1.54, before any gain from the enlarged pool or from partial hypotheses.

6.2 Phase 0: measured ingredients

Everything the fit is parameterised by has been measured on truth-matched PFPs and recorded in `HypFitParams.h` (700 files, 1323 signal events).

constraint	n	Gaussian centre	σ
$m(p,\pi)$ vs Λ (1.11568)	397	1.1234	27.2 MeV
$m(\Lambda,\gamma)$ vs Σ^0 (1.19264)	243	1.1812	45.4 MeV

Table 5: Mass constraints from correctly assigned candidates. The Σ^0 term is $\sim 1.7\times$ wider, as expected from the soft photon.

The reconstruction is biased by +7.7 MeV on the Λ and −11.4 MeV on the Σ^0, which is 28% and 25% of the respective widths. **The χ^2 must therefore be centred on the measured means, not the PDG masses**, or every hypothesis carries a built-in offset. Both biases have identifiable causes: the pion momentum is taken from range under the *muon* hypothesis, and the photon energy is under-measured (reco 0.0792 vs true 0.0861 GeV).

Remaining ingredients:

- p–π distance of closest approach: median 0.35 cm, 16–84% range 0.08–1.36 cm. Strongly peaked and one of the most powerful terms available — and note that it works precisely where the Λ decay *distance* fails.
- The muon is the longest of the three tracks 92.1% of the time: a soft prior worth $\ln(0.921/0.079) = 2.46$ in log-likelihood, not a cut.
- 98.9% of PFPs carry *both* a track fit and a shower fit, so the relaxed pool requires no re-reconstruction — only a change of eligibility rule.

variable	μ^+	p	π^-	γ
`trk_shr_score`	1.00	0.90	0.80	**0.11**
`trk_llrpid`	**0.96**	−0.29	0.14	0.56
`trk_pid_chipr`	180	**16.4**	96	192
`three_plane_dedx`	2.47	**10.3**	7.52	3.30
`rms_deflection`	0.01	0.01	0.01	**0.08**
`moliere_avg`	1.78	6.18	8.95	**13.3**
`trk_length`	**139**	12.8	10.8	14.2

Table 6: Per-role PID templates, medians on truth-matched PFPs ($n = 1213/808/650/814$). The muon, proton and photon each have a distinctive signature; the π^- has none.

6.3 The mass-sculpting risk

Selecting the combination that best fits m_Λ will sculpt a Λ-like peak out of pure combinatorics in background events, manufacturing the signal being searched for. This is the standard failure mode of combinatoric selection and it is the main risk in the whole scheme.

Decision: the Λ and Σ^0 invariant masses are used for assignment only and are kept out of the BDT input list. The best-hypothesis score and the margin to the second-best hypothesis are permitted as inputs. Before the fit is adopted, the full machinery must be run on a background-only sample and the resulting mass distributions inspected for an induced peak. A secondary check is to assign using one constraint and test with the other.

A second, smaller risk: the enlarged candidate pool raises background acceptance as well as signal efficiency, so the figure of merit must be re-measured end to end rather than judged on efficiency alone.

7 Fake data by Poisson throw

The problem. The current "data" sample is the signal file itself (`SigmaSelection.C` maps the same NuWro hyperon ntuple to both sample 0 and sample 4), scaled by POT. That is not a closure test: it contains no background, no statistical fluctuation, and is perfectly correlated with the signal prediction it would be compared against.

The fix. Adopt the same Poisson-throw scheme the CC1μ1π analysis uses (`macros/throw_perrun_*.C`). For each MC event with central-value weight w_{cv} and POT scale $\alpha = \mathrm{POT}_{\mathrm{data}}/\mathrm{POT}_{\mathrm{MC}}$, draw

$$n \sim \mathrm{Poisson}(w_{\mathrm{cv}}\,\alpha) \tag{4}$$

and write the event n times into the fake-data tree with all weights reset to unity. Applied across *all* samples – signal, other hyperons, background, dirt – this produces a fake dataset that

- contains the correct background admixture, which matters enormously here because background outnumbers signal by $\sim$900:1;
- carries the right statistical fluctuations, so uncertainties can be validated rather than assumed;
- is statistically independent of the prediction, making a genuine closure test possible.

Implementation. A `throw_sigma0.C` modelled on `throw_perrun_w.C`: chain the processed MC, drop the multisim weight branches, `CloneTree(0)` to preserve the Σ^0 observables, throw, and stamp `summed_pot` with the target POT. One seed per sample so the throw is reproducible.

Caveat to keep in view. A Poisson throw from the same MC is a *statistical* closure test, not a model test. It cannot reveal generator mismodelling, because the "data" and the prediction come from the same generator. A genuine model test needs a hyperon sample from a different generator – GENIE against the current NuWro. One now exists in *hypana* format (`lists/sig_genie.list`, 4851 files, 5.9770×10^{22} POT) but not in PeLEE format, which is what the fake-data machinery consumes; a GENIE overlay production is running locally and would close the gap. Until it does, the throw validates the statistical and normalisation chain only, and that limitation should be stated wherever the closure is quoted. Sec. 8 shows the two generators differ by roughly a factor of four on the Σ^0 rate, so this is not a small test to be skipped.

8 The signal sample and its normalisation

8.1 The run3b production

The Run 1-EXT sample `prod_hyperon_sigmazero_reco1_prodnuwro_mcc9` was retired on 2026-10-15. Its replacement is produced locally (`~/MicroBooNE/production/nuwro_prod/`): 2500 jobs of 100 NuWro Σ^0 events overlaid on swizzled NuMI **run3b ext_unbiased** data, through the full gen$\rightarrow$reco2 chain, then both hypana and PeLEE ntuples.

	retired (Run 1 EXT)	run3b
files / events	2981 / 64716	2500 / 249832
generated POT	3.4141×10^{23}	1.3394×10^{24}
true Σ^0 CC $\bar{\nu}_\mu$	17714 (27.4%)	58545 (23.5%)
raw signal at preselection	3226	10931

The hypana `ana/MetaTree` sum and the merged PeLEE `SubRun` sum agree on the POT to four digits, and `check_ext_unique.py` confirms 250,000 distinct EXT events with none used twice. The cosmics now come from the same run period as the overlay background, which the retired sample did not: it was Run 1 EXT under a Run 1+2+4 background. Selection efficiency is unchanged by the swap (50.1% → 50.8% at preselection), so the gain is purely statistical.

8.2 The NuMI flux weight

NuWro's flux files were converted from the *old* NuMI gsimple flux (g4numi v6, bugged beamline geometry) while the overlay, the dirt and the data are on the current `g4_10_4` flux. Every NuWro event therefore carries the official MicroBooNE "New/Old" ratio weight, binned in $(E_\nu, \theta_{\text{beam}})$. Measured over the finished production the mean weight is 0.6774, and 0.6770 **for the Σ^0 CC subset** – flat to 0.1% even though those events sit 0.13 GeV higher in energy than the rest. There is no hyperon-specific correction to make, only a wider spread (rms 0.297 against 0.226) costing a little effective statistics, 49.9k of 58.5k. `SigmaSelection.C` computes the weight on the fly from `nu_e` and `true_nu_p*`; `HypCutFlow.C` reads a per-event table.

8.3 Why the samples disagree on the absolute Σ^0 rate

On one truth definition and one fiducial volume, Σ^0 CC $\bar{\nu}_\mu$ events per 1.1526×10^{21} POT with the true vertex in the FV:

sample	in-FV Σ^0	vs the plain overlay
plain RHC overlay (unfiltered GENIE)	83.3	1.00
GENIE hyperon sample `sig_genie`	110.1	1.32
NuWro run3b, POT scale only	18.9	0.23
NuWro run3b, with the flux weight	12.8	**0.15**

The fiducial fractions are 37–38% in all three, so this is not a generation-volume artefact.

It is hyperon FSI. The NuWro configuration (`ultrapure-sigma0-numi-rhc.txt`) sets `dyn_hyp_cc=1` with every other channel off and, within it, `hyp_lambda=0`, `hyp_sigma_zero=1`, `hyp_sigma_minus=0`: NuWro generated the Σ^0 channel *and nothing else*, so its quoted POT per event is the exposure per Σ^0 *produced* – exactly the normalisation a Σ^0 signal sample wants. Everything else in the sample is final-state interaction: of the $\bar{\nu}_\mu$ CC events only 25.3% still carry a Σ^0, 27.9% carry a Λ ($\Sigma^0 N \to \Lambda N$ in the cascade), 10% a $\Sigma^\pm$, and 37.3% no strange particle at all. The MicroBooNE Λ paper (arXiv:2212.07888) states the other half outright: "the NuWro predictions include the effect of final state interactions, while GENIE does not take them into account". For GENIE, produced = surviving, so the comparison that means anything is Σ^0 *produced*:

	in-FV Σ^0 produced per full-RHC POT
GENIE overlay (no hyperon FSI)	83.3
NuWro, POT scale only	79.1
NuWro, with the flux weight	53.6

The 6.5× gap decomposes exactly as 4.19× (FSI) × 1.48× (flux weight) × 1.05× (production rate), product 6.51× against the measured 6.51×. The sign flip between channels is the same effect: in Fig. 2 of the Λ paper NuWro sits *above* GENIE ($\simeq 2.1$ against $\simeq 1.5 \times 10^{-40}$ cm^2/Ar) because FSI *adds* Λ, and far below for Σ^0 because it *removes* Σ^0.

A control run verifies the normalisation. The same flux, target and configuration with `dyn_hyp_cc=0 dyn_qel_cc=1` – ordinary CC quasi-elastic instead of the hyperon channel – gives 7.539×10^{15} POT per event over NuWro's box. The overlay's own CC QEL rate (20,277 in-FV events in 7.1479×10^{20} POT) converts to 8.835×10^{15} on the same footing, using the 37.0% box-mass fiducial fraction and the 0.677 flux factor. NuWro's generic event rate is therefore within 17% of GENIE's, which is the ordinary CCQE model difference, and its flux × target normalisation is sound. Referenced to each generator's own CCQE rate, NuWro's Σ^0 *production* is 0.81× GENIE's.

What this means for the limit. A Σ^0 yield taken from the NuWro sample is a yield *with* hyperon FSI; one taken from the GENIE samples is a yield *without* it. The difference is roughly a factor of four and it is a model choice, not a bug to be fixed. The measured Λ cross section, $2.0^{+2.2}_{-1.7} \times 10^{-40}$ cm^2/Ar, brackets both generators and cannot currently distinguish them. This has to be carried as a signal-model systematic on any limit, and it is larger than anything in the detector or flux budget. The residual 1.3–1.6× among the GENIE samples themselves – the same factor the Λ cross-check reports as 1.61× – is unexplained and separate.

9 Exposure: RHC now, combined later

Now: RHC only. Everything is scaled to the full RHC data exposure, $\mathrm{POT}_{\mathrm{data}} = 1.1526 \times 10^{21}$:

Sample	Source	Generated POT
Σ^0 signal / hyperons	NuWro run3b, ×0.677 flux weight	1.3394×10^{24}
Background	RHC overlay, Runs 1+2+4a+4b+4c	9.94482×10^{21}
Dirt	Run 1 RHC only	4.21274×10^{20}

One known imperfection remains: the dirt sample is Run-1 RHC only, scaled up to the full RHC exposure. The second – a signal sample generated with no cosmic overlay, whose non-muon truth matching could not be trusted – is gone: the run3b sample is overlaid on real run3b EXT data, the same period as the background. The signal POT above is the generated POT; the 0.677 flux weight is applied per event, not folded into it (Sec. 8).

Later: combined FHC+RHC. The signal is $\bar{\nu}_\mu$ charged-current, so RHC is the sensitive mode – but FHC carries a substantial wrong-sign $\bar{\nu}_\mu$ component (34.0% of the FHC flux, against 53.3% for RHC), and the FHC exposure is 8.857×10^{20} POT. Folding FHC in therefore adds real signal rather than only background, and the CC1μ1π analysis already has the machinery: per-run POT scaling, a combined `file_properties`, and combined flux normalisation.

Deferred deliberately. The combination is a normalisation exercise, and at 0.1% purity it would add background faster than signal in absolute terms. The selection work of Sec. 4 comes first; the exposure extension is worth doing once the selection is worth extending. When it happens it needs: FHC overlay samples processed through the same selection, an FHC hyperon signal sample (or a justified reuse of the RHC one, given the signal is beam-mode independent at the interaction level), FHC dirt, and a combined-mode flux normalisation following `Flux` in the CCπ xsec configs.

10 Work plan

1. **Selection (priority): global hypothesis fit.** Relaxed candidate pool, combinatoric role assignment scored against the Λ and Σ^0 mass constraints and the per-role PID templates, then partial hypotheses for three-PFP events. Sec. 6. Phase 0 (resolutions, templates, priors) is complete and recorded in `HypFitParams.h`; Sec. 6.2. *The Λ displaced-vertex requirement listed in earlier versions of this plan is withdrawn: the true flight distance (median 1.85 cm) is three times smaller than the vertex resolution (5.54 cm) and the reconstructed quantity is uncorrelated with truth, $r = -0.014$. Sec. 5.*

2. **Framework port.** Branch bindings, `Sigma0` selection class, factory registration, cut-flow validation against the standalone code before any config work. *Not started as of 2026-10-15.* Sec. 2.

3. **Background through HyperonProduction.** *Unblocked.* The RHC overlay has been transferred, run through reco2 and processed with HyperonProduction: 14,024 hypana files at 6.9701×10^{21} POT, 70% of the full RHC exposure. A hypana-only analysis with realistic background is now possible; dirt in hypana format is still missing. Sec. 5.

4. **Fake data.** `throw_sigma0.C`, all samples, RHC POT. Sec. 7.

5. **Signal-model systematic.** The hyperon-FSI difference between NuWro and GENIE is a factor ~ 4 on the Σ^0 yield (Sec. 8) and dominates every other uncertainty. It has to be evaluated and carried before a limit means anything.

6. **Systematics.** Single-bin `univmake` for the background covariance.

7. **Combined exposure.** FHC+RHC once the selection justifies it. Sec. 9.

8. **Limit.** Outside the framework, reading its covariance.